1 Pseudocode of MSFS

Algorithm 1 MSFS

Training dataset $\mathcal{D}$, number of clusters m, stretching ratio α
Fine-tuned model with updated embeddings

procedure EmbeddingStretching($\mathcal{D}, m, \alpha$)
 $\mathcal{Z} \leftarrow \{f_e(x) \mid x \in \mathcal{D}\}$ ▷ Get the embeddings of all samples
 $\{C_1, \ldots, C_m\}, \mathcal{C} \leftarrow$ K-Means($\mathcal{Z}, m$) ▷ Cluster embeddings
 $\mathbf{s} \leftarrow \frac{1}{|\mathcal{C}|} \sum_{c \in \mathcal{C}} c$ ▷ Compute stretching core
 $\mathcal{E} \leftarrow \emptyset$ ▷ Original set of excess samples
 for each cluster C_i **do**
  Compute label counts in C_i
  $n_{\min} \leftarrow \min_{y \in Y} \text{count}(y)$
  for each label y in C_i **do**
   if count(y) $> n_{\min}$ **then**
    Add excess samples of label y to $\mathcal{E}$
   end if
  end for
 end for
 ▷ Fine-tuning with embedding modification
 for each batch (x, y) in $\mathcal{D}$ **do**
  $z \leftarrow f_e(x)$ ▷ Original embedding
  if $x \in \mathcal{E}$ **then**
   $z' \leftarrow \mathbf{s} + \alpha(z - \mathbf{s})$ ▷ Stretch toward sc
  else
   $z' \leftarrow z$
  end if
  Forward z' through remaining layers
  Compute loss and update model
 end for
end procedure

2 What is 'Stretching'

In this paper, 'stretching' refers to the transformation of an embedding tensor in the embedding space. However, since all transformations applied to the embedding tensor in this study are relative transformations of a point with respect to a fixed point, which alter the distance between the point and the stretching core, 'stretching' in fact denotes the change in the value of the embedding tensor at that point resulting from the modification of its relative distance to the stretching core.

3 How to Determine an Appropriate Stretching Ratio

In this section, we explore how to identify an appropriate stretching ratio for the MSFS method.

Observing Appendix Tables 5 and 6 and examining various types of spurious correlations across three datasets, we find that for Occur-type and concept-type spurious correlations, MSFS more readily demonstrates enhanced spurious correlation mitigation at larger stretching ratios, such as 100, 1000, or 10000. We posit that this is because Occur- and concept-type spurious correlations primarily arise from high frequencies of word or phrase occurrences, and direct, substantial embedding stretching can directly differentiate embedding semantics among words to decouple such spurious correlations. In contrast, the optimal stretching ratio for Style-type spurious correlations generally lies at smaller values, such as 0.001 or 0.1. We attribute this to the abstract nature of Style-type spurious correlations, where conceptual semantic meanings of texts require joint expression by multiple words; thus, aggregating such embeddings within a concentrated region of the embedding space may better facilitate differentiation of their conceptual distinctions.

The Yelp dataset, with its large size, can also support effective fitting of stretched sample features under substantial embedding stretching. In contrast, the Emotion and Beer datasets, being smaller, often exhibit optimal stretching ratios at settings of 0.1 and 10. When the embedding stretching magnitude is too large, the smaller data size hinders effective learning of overall features.

4 Additional analysis of the Main Results

Observing the model's accuracy on the normal-test set and anti-test set reveals that, under the optimal stretching ratio, MSFS effectively improved the accuracy on the anti-test set without sacrificing accuracy on the normal-test set, with substantial improvements on some datasets. For example, the anti-test accuracy on Emotion:Catg reached 75.23%, significantly outperforming existing spurious correlation mitigation approaches. This demonstrates the effectiveness of MSFS in enhancing the model's robustness to spurious correlations.

5 Setups

We selected the dataset version with $\lambda = 1$, the version with the highest spurious correlation intensity, and split training and validation sets in a 9:1 ratio. Experiments were conducted on all 12 datasets of Shortcuts Maze using all selected baselines and MSFS. Each experiment ran for ten rounds, selecting the optimal model for test set evaluation.

To ensure the validity of Comparison Δ, we manually removed data with failed or nearly failed performances on the normal-test set, along with corresponding performances on the anti-test set. Testing utilized NVIDIA V100 GPUs with a fixed batch size of 32. Method stability was assessed via statistical significance testing with three random seeds.

For fairness, baseline default settings were maintained except JTT's and BAM's parameter λ_{up} set to 3 to prevent overfitting on small datasets. For MSFS, hyperparameter m was fixed at 10, and only the stretching ratio—the hyperparameter with the greatest impact on results—underwent grid search to identify the optimal value. [1]

6 Preliminaries

We consider a text classification task in natural language processing, where each sample in the dataset comprises text content and a task label, but lacks traditional spurious correlation group labels.

Spurious correlations are implicitly expressed and confounded within the sample texts of the dataset. Let the training set be denoted as $\mathcal{D}_{\text{train}} = \{(x, y) \mid x \in \mathcal{X}, y \in \mathcal{Y}\}$, where x represents a sample text in the input space $\mathcal{X}$ and y denotes the corresponding task label in the finite label space $\mathcal{Y}$.

Our goal is to learn a model $f_\theta : \mathcal{X} \to \mathcal{Y}$, parameterized by θ from the parameter space Θ, that predicts the task label of sample texts. Specifically, we aim to find a set of model parameters $\theta^* \in \Theta$ that minimizes the discrepancy in prediction accuracy between the normal-test set $\mathcal{D}_{\text{test}}^{\text{norm}}$ and the anti-test set $\mathcal{D}_{\text{test}}^{\text{anti}}$, formulated as:

$$\theta^* = \arg\min_{\theta \in \Theta} \left| \text{Acc}(f_\theta; \mathcal{D}_{\text{test}}^{\text{norm}}) - \text{Acc}(f_\theta; \mathcal{D}_{\text{test}}^{\text{anti}}) \right|, \tag{1}$$

where $\text{Acc}(f_\theta; \mathcal{D}) = \frac{1}{|\mathcal{D}|} \sum_{(x,y)\in\mathcal{D}} \mathbb{I}(f_\theta(x) = y)$ denotes the prediction accuracy of model f_θ on dataset $\mathcal{D}$, and $\mathbb{I}(\cdot)$ is the indicator function.

To evaluate the robustness of the method against spurious correlation issues, we employ the Shortcut Maze dataset for assessment.

7 Baselines

We selected currently prevalent spurious correlation mitigation approaches that can be directly applied to text classification tasks, are suitable for Shortcuts Maze, and require no group label-based training or evaluation. These include two-stage strategies, namely JTT [4], BAM [3], and EA [7], and loss function and regularization term optimization approaches, specifically NFL [1] and ER [6]. Additionally, we included the conventional ERM [5] loss function as one of the comparison baselines. We uniformly employed BERT [2] as the backbone model to conduct experiments on the aforementioned methods.

[1] The stretching ratios used in this grid search were 0.001, 0.01, 0.1, 10, 100, 1000, and 10000, respectively.

Model	Dataset Accuracy (%) Performance											
	Yelp					Emotion					Beer	
	Occur			Style		Occur			Style		Concept	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
ERM	$64.42_{\pm 0.34}$	$63.75_{\pm 0.36}$	$64.45_{\pm 0.31}$	$61.02_{\pm 0.83}$	$60.80_{\pm 0.97}$	$90.41_{\pm 0.48}$	$90.66_{\pm 0.98}$	$90.75_{\pm 0.73}$	$88.37_{\pm 0.68}$	$76.55_{\pm 3.79}$	$76.90_{\pm 0.39}$	$89.57_{\pm 0.40}$
EA	$61.37_{\pm 0.09}$	$61.50_{\pm 0.40}$	$62.15_{\pm 0.12}$	$57.61_{\pm 0.77}$	$57.05_{\pm 0.56}$	$89.05_{\pm 0.43}$	$89.15_{\pm 1.07}$	$89.25_{\pm 0.30}$	$83.99_{\pm 2.94}$	$73.19_{\pm 2.74}$	$75.23_{\pm 1.07}$	$87.32_{\pm 1.57}$
JTT	$60.28_{\pm 0.25}$	$60.42_{\pm 0.26}$	$60.92_{\pm 0.16}$	$57.37_{\pm 0.41}$	$57.53_{\pm 0.16}$	$91.68_{\pm 0.93}$	$91.53_{\pm 0.78}$	$89.93_{\pm 0.24}$	$90.32_{\pm 0.42}$	$75.52_{\pm 0.59}$	$77.37_{\pm 0.19}$	$88.80_{\pm 0.23}$
NFL-CO	$64.11_{\pm 0.08}$	$63.51_{\pm 0.36}$	$64.19_{\pm 0.56}$	$61.08_{\pm 0.21}$	$61.22_{\pm 0.44}$	$86.08_{\pm 1.55}$	$84.14_{\pm 4.32}$	$88.47_{\pm 0.66}$	$86.04_{\pm 1.59}$	$77.03_{\pm 2.24}$	$71.38_{\pm 3.03}$	$83.92_{\pm 0.49}$
BAM	$59.25_{\pm 0.16}$	$59.64_{\pm 0.41}$	$59.96_{\pm 0.22}$	$56.43_{\pm 0.24}$	$57.38_{\pm 0.20}$	$90.95_{\pm 0.55}$	$90.22_{\pm 0.54}$	$90.07_{\pm 0.74}$	$88.32_{\pm 0.21}$	$74.65_{\pm 0.37}$	$74.70_{\pm 0.35}$	$87.20_{\pm 0.25}$
ER	$63.35_{\pm 0.84}$	$63.29_{\pm 0.24}$	$63.78_{\pm 0.23}$	$61.08_{\pm 0.21}$	$60.74_{\pm 0.37}$	$86.47_{\pm 2.73}$	$84.23_{\pm 2.82}$	$87.69_{\pm 3.68}$	$85.30_{\pm 1.37}$	$75.77_{\pm 2.38}$	$74.17_{\pm 0.70}$	$87.12_{\pm 0.86}$

Table 1: Accuracy Statistical Table of Baselines on Shortcuts Maze Normal-Test

Model	Dataset Accuracy (%) Performance											
	Yelp					Emotion					Beer	
	Occur			Style		Occur			Style		Concept	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
ERM	$37.47_{\pm 0.96}$	$42.00_{\pm 1.76}$	$42.56_{\pm 0.58}$	$43.83_{\pm 0.42}$	$35.54_{\pm 0.13}$	$21.90_{\pm 1.24}$	$35.18_{\pm 10.19}$	$43.80_{\pm 11.28}$	$29.68_{\pm 1.16}$	$18.93_{\pm 1.54}$	$63.50_{\pm 3.11}$	$64.12_{\pm 5.58}$
EA	$37.27_{\pm 1.30}$	$45.39_{\pm 1.17}$	$39.30_{\pm 2.08}$	$37.01_{\pm 1.68}$	$33.26_{\pm 2.03}$	$20.58_{\pm 1.02}$	$45.74_{\pm 6.46}$	$45.40_{\pm 10.31}$	$27.40_{\pm 2.82}$	$15.82_{\pm 1.07}$	$61.03_{\pm 1.43}$	$67.57_{\pm 3.40}$
JTT	$41.57_{\pm 0.56}$	$45.64_{\pm 0.00}$	$35.51_{\pm 1.45}$	$39.77_{\pm 0.26}$	$31.66_{\pm 0.18}$	$24.62_{\pm 1.67}$	$29.05_{\pm 1.21}$	$42.04_{\pm 7.73}$	$31.53_{\pm 0.36}$	$21.07_{\pm 2.41}$	$60.25_{\pm 0.63}$	$50.03_{\pm 1.69}$
NFL-CO	$43.01_{\pm 2.70}$	$49.05_{\pm 2.71}$	$50.92_{\pm 4.18}$	$43.16_{\pm 1.32}$	$35.41_{\pm 1.57}$	$22.28_{\pm 2.41}$	$44.87_{\pm 12.20}$	$58.30_{\pm 5.40}$	$28.42_{\pm 1.71}$	$13.23_{\pm 1.07}$	$58.15_{\pm 6.30}$	$68.72_{\pm 4.32}$
BAM	$41.52_{\pm 0.31}$	$46.23_{\pm 0.31}$	$37.30_{\pm 0.42}$	$40.71_{\pm 0.20}$	$33.23_{\pm 0.52}$	$23.80_{\pm 2.09}$	$33.87_{\pm 2.79}$	$44.28_{\pm 6.24}$	$31.29_{\pm 2.12}$	$20.68_{\pm 1.08}$	$60.13_{\pm 1.09}$	$50.27_{\pm 0.54}$
ER	$39.28_{\pm 2.37}$	$45.92_{\pm 2.20}$	$45.77_{\pm 2.15}$	$41.01_{\pm 0.21}$	$35.19_{\pm 1.33}$	$17.27_{\pm 1.49}$	$43.99_{\pm 8.23}$	$43.50_{\pm 3.71}$	$29.68_{\pm 1.67}$	$15.23_{\pm 1.08}$	$62.97_{\pm 0.79}$	$66.08_{\pm 1.96}$

Table 2: Accuracy Statistical Table of Baselines on Shortcuts Maze Anti-Test

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Model	Dataset F1 (%) Performance											
	Yelp					Emotion					Beer	
	Occur			Style		Occur			Style		Concept	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
ERM	$64.28_{\pm 0.30}$	$63.73_{\pm 0.48}$	$64.19_{\pm 0.55}$	$61.20_{\pm 0.87}$	$60.24_{\pm 0.66}$	$82.66_{\pm 0.28}$	$82.03_{\pm 1.38}$	$83.52_{\pm 0.48}$	$78.46_{\pm 1.22}$	$60.56_{\pm 3.62}$	$76.48_{\pm 0.28}$	$88.82_{\pm 0.48}$
EA	$61.59_{\pm 0.62}$	$60.17_{\pm 0.68}$	$61.21_{\pm 0.88}$	$56.02_{\pm 1.26}$	$54.12_{\pm 1.44}$	$69.57_{\pm 7.47}$	$73.60_{\pm 2.22}$	$81.71_{\pm 0.92}$	$70.79_{\pm 2.74}$	$51.32_{\pm 5.68}$	$74.84_{\pm 0.74}$	$86.76_{\pm 1.61}$
JTT	$60.43_{\pm 0.24}$	$60.65_{\pm 0.26}$	$61.23_{\pm 0.15}$	$57.78_{\pm 0.48}$	$57.86_{\pm 0.17}$	$84.80_{\pm 1.74}$	$84.51_{\pm 1.01}$	$82.89_{\pm 0.83}$	$82.98_{\pm 0.51}$	$64.40_{\pm 0.72}$	$77.16_{\pm 0.19}$	$87.97_{\pm 0.17}$
NFL-CO	$63.91_{\pm 0.14}$	$63.21_{\pm 0.36}$	$63.83_{\pm 0.72}$	$60.66_{\pm 0.17}$	$60.61_{\pm 1.06}$	$73.84_{\pm 2.07}$	$71.78_{\pm 4.66}$	$77.93_{\pm 1.11}$	$74.87_{\pm 1.51}$	$62.72_{\pm 1.52}$	$71.52_{\pm 3.15}$	$83.74_{\pm 0.51}$
BAM	$59.21_{\pm 0.22}$	$59.77_{\pm 0.42}$	$60.25_{\pm 0.23}$	$56.68_{\pm 0.23}$	$57.59_{\pm 0.20}$	$83.16_{\pm 1.82}$	$81.80_{\pm 1.05}$	$82.49_{\pm 0.61}$	$80.53_{\pm 0.46}$	$64.02_{\pm 0.57}$	$74.50_{\pm 0.27}$	$86.38_{\pm 0.16}$
ER	$62.48_{\pm 1.43}$	$63.11_{\pm 0.24}$	$63.16_{\pm 0.55}$	$60.82_{\pm 0.14}$	$60.08_{\pm 0.55}$	$67.68_{\pm 8.93}$	$63.77_{\pm 7.64}$	$73.49_{\pm 12.44}$	$66.23_{\pm 7.51}$	$60.94_{\pm 3.47}$	$73.71_{\pm 0.78}$	$86.48_{\pm 0.71}$

Table 3: F1 Statistical Table of Baselines on Shortcuts Maze Normal-Test

Model	Dataset F1 (%) Performance											
	Yelp					Emotion					Beer	
	Occur			Style		Occur			Style		Concept	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
ERM	$33.19_{\pm 2.48}$	$39.17_{\pm 2.65}$	$39.79_{\pm 1.17}$	$42.39_{\pm 0.43}$	$30.62_{\pm 0.19}$	$29.20_{\pm 1.98}$	$41.28_{\pm 6.04}$	$48.86_{\pm 6.13}$	$30.75_{\pm 1.70}$	$17.97_{\pm 1.13}$	$62.71_{\pm 4.21}$	$64.39_{\pm 5.22}$
EA	$32.71_{\pm 1.97}$	$44.49_{\pm 1.33}$	$35.72_{\pm 3.59}$	$32.44_{\pm 2.38}$	$25.66_{\pm 4.53}$	$27.85_{\pm 5.00}$	$44.99_{\pm 4.69}$	$49.88_{\pm 5.07}$	$33.32_{\pm 1.37}$	$18.19_{\pm 5.09}$	$59.26_{\pm 0.43}$	$68.36_{\pm 3.29}$
JTT	$39.63_{\pm 0.78}$	$44.96_{\pm 0.08}$	$30.44_{\pm 2.26}$	$38.17_{\pm 0.37}$	$25.91_{\pm 0.28}$	$33.34_{\pm 1.06}$	$41.87_{\pm 0.50}$	$44.84_{\pm 5.29}$	$32.48_{\pm 1.07}$	$19.06_{\pm 1.90}$	$58.48_{\pm 0.65}$	$52.04_{\pm 1.64}$
NFL-CO	$40.70_{\pm 3.98}$	$49.07_{\pm 3.23}$	$50.58_{\pm 5.25}$	$42.22_{\pm 1.81}$	$30.16_{\pm 2.84}$	$31.15_{\pm 1.91}$	$42.13_{\pm 6.11}$	$52.78_{\pm 5.22}$	$28.48_{\pm 1.44}$	$16.59_{\pm 2.55}$	$58.25_{\pm 6.25}$	$69.60_{\pm 3.39}$
BAM	$40.49_{\pm 0.43}$	$46.27_{\pm 0.36}$	$34.44_{\pm 0.75}$	$39.82_{\pm 0.32}$	$28.01_{\pm 0.69}$	$32.35_{\pm 1.54}$	$39.93_{\pm 0.58}$	$45.70_{\pm 4.71}$	$31.44_{\pm 1.88}$	$20.00_{\pm 0.43}$	$59.32_{\pm 1.17}$	$52.12_{\pm 0.39}$
ER	$36.37_{\pm 4.38}$	$45.36_{\pm 2.83}$	$44.66_{\pm 3.31}$	$39.28_{\pm 0.18}$	$29.85_{\pm 1.56}$	$22.41_{\pm 2.08}$	$35.91_{\pm 3.73}$	$41.13_{\pm 7.76}$	$24.85_{\pm 0.60}$	$14.19_{\pm 0.69}$	$61.82_{\pm 1.43}$	$66.44_{\pm 1.85}$

Table 4: F1 Statistical Table of Baselines on Shortcuts Maze Anti-Test

Model	Dataset Accuracy (%) Performance											
	Yelp					Emotion					Beer	
	Occur			Style		Occur			Style		Concept	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
0.001	$61.89_{\pm 0.28}$	$62.04_{\pm 0.28}$	$62.07_{\pm 0.05}$	$\mathbf{57.02_{\pm 0.98}}$	$58.01_{\pm 1.28}$	$84.43_{\pm 0.18}$	$87.69_{\pm 1.76}$	$88.37_{\pm 2.52}$	$\mathbf{71.53_{\pm 4.96}}$	$68.47_{\pm 2.42}$	$63.33_{\pm 0.74}$	$85.00_{\pm 1.07}$
0.01	$62.58_{\pm 0.16}$	$62.98_{\pm 0.04}$	$62.46_{\pm 0.84}$	$59.44_{\pm 0.47}$	$\mathbf{58.62_{\pm 0.26}}$	$87.20_{\pm 0.59}$	$86.57_{\pm 1.78}$	$90.12_{\pm 0.59}$	$78.64_{\pm 4.04}$	$\mathbf{69.05_{\pm 4.65}}$	$67.02_{\pm 1.58}$	$87.22_{\pm 0.43}$
0.1	$63.56_{\pm 0.74}$	$63.47_{\pm 0.07}$	$63.39_{\pm 0.54}$	$60.72_{\pm 0.55}$	$60.92_{\pm 0.19}$	$91.00_{\pm 1.24}$	$90.56_{\pm 0.55}$	$92.46_{\pm 0.42}$	$90.17_{\pm 0.69}$	$77.86_{\pm 1.31}$	$76.80_{\pm 0.69}$	$\mathbf{87.98_{\pm 1.00}}$
10.0	$64.21_{\pm 0.75}$	$63.79_{\pm 0.17}$	$63.46_{\pm 0.69}$	$59.69_{\pm 1.38}$	$59.65_{\pm 0.17}$	$90.61_{\pm 1.08}$	$91.24_{\pm 0.83}$	$90.85_{\pm 0.18}$	$87.74_{\pm 0.78}$	$73.09_{\pm 2.91}$	$76.47_{\pm 0.91}$	$89.02_{\pm 1.27}$
100.0	$\mathbf{63.57_{\pm 0.11}}$	$63.17_{\pm 0.22}$	$62.91_{\pm 0.12}$	$58.78_{\pm 0.87}$	$57.67_{\pm 0.28}$	$\mathbf{89.15_{\pm 1.11}}$	$\mathbf{85.79_{\pm 2.73}}$	$\mathbf{91.19_{\pm 0.25}}$	$88.52_{\pm 0.79}$	$75.13_{\pm 1.90}$	$\mathbf{74.62_{\pm 0.80}}$	$86.67_{\pm 1.07}$
1000.0	$63.49_{\pm 0.41}$	$\mathbf{62.33_{\pm 0.50}}$	$62.49_{\pm 0.44}$	$56.21_{\pm 2.91}$	$56.77_{\pm 0.91}$	$88.42_{\pm 0.91}$	$86.37_{\pm 0.61}$	$91.68_{\pm 0.73}$	$87.69_{\pm 1.38}$	$72.55_{\pm 2.91}$	$74.32_{\pm 0.16}$	$85.93_{\pm 1.58}$
10000.0	$61.94_{\pm 1.31}$	$62.39_{\pm 0.55}$	$\mathbf{61.49_{\pm 0.60}}$	$56.74_{\pm 1.56}$	$55.91_{\pm 0.26}$	$87.59_{\pm 1.56}$	$84.87_{\pm 2.94}$	$91.44_{\pm 0.73}$	$86.23_{\pm 1.74}$	$73.58_{\pm 3.73}$	$73.90_{\pm 0.66}$	$85.52_{\pm 2.15}$

Table 5: **Accuracy Statistical Table of MSFS on Shortcuts Maze Normal-Test.** The result of the optimal stretching ratio is marked in bold.

Model	Dataset Accuracy (%) Performance											
	Yelp					Emotion					Beer	
	Occur			Style		Occur			Style		Concept	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
0.001	$35.53_{\pm 0.97}$	$46.45_{\pm 0.25}$	$44.91_{\pm 1.01}$	$\mathbf{45.41_{\pm 0.79}}$	$34.96_{\pm 0.26}$	$18.05_{\pm 1.37}$	$37.13_{\pm 7.26}$	$33.04_{\pm 7.75}$	$\mathbf{26.72_{\pm 4.99}}$	$16.84_{\pm 1.52}$	$52.83_{\pm 4.34}$	$57.58_{\pm 4.09}$
0.01	$37.30_{\pm 1.01}$	$44.93_{\pm 1.69}$	$42.72_{\pm 1.40}$	$45.23_{\pm 0.12}$	$\mathbf{35.71_{\pm 0.12}}$	$18.39_{\pm 1.49}$	$37.86_{\pm 10.51}$	$44.04_{\pm 9.24}$	$32.46_{\pm 2.57}$	$\mathbf{17.81_{\pm 1.58}}$	$54.33_{\pm 2.52}$	$63.67_{\pm 5.59}$
0.1	$39.35_{\pm 1.64}$	$46.20_{\pm 1.01}$	$47.48_{\pm 0.98}$	$44.50_{\pm 0.87}$	$35.82_{\pm 0.79}$	$21.95_{\pm 2.15}$	$29.78_{\pm 3.32}$	$48.32_{\pm 10.87}$	$33.97_{\pm 2.15}$	$19.08_{\pm 1.55}$	$66.17_{\pm 0.55}$	$\mathbf{65.98_{\pm 3.33}}$
10.0	$40.56_{\pm 1.01}$	$45.60_{\pm 2.16}$	$43.73_{\pm 1.18}$	$36.21_{\pm 5.13}$	$34.07_{\pm 1.58}$	$22.24_{\pm 0.89}$	$34.16_{\pm 3.42}$	$49.54_{\pm 12.53}$	$30.66_{\pm 1.29}$	$15.72_{\pm 1.74}$	$65.05_{\pm 3.03}$	$61.68_{\pm 1.33}$
100.0	$\mathbf{43.68_{\pm 1.87}}$	$48.13_{\pm 1.41}$	$47.12_{\pm 1.47}$	$37.53_{\pm 1.39}$	$27.38_{\pm 2.71}$	$\mathbf{24.18_{\pm 6.56}}$	$\mathbf{43.94_{\pm 13.09}}$	$\mathbf{75.23_{\pm 1.92}}$	$28.95_{\pm 2.68}$	$13.04_{\pm 0.88}$	$\mathbf{64.13_{\pm 1.18}}$	$60.80_{\pm 5.64}$
1000.0	$40.55_{\pm 1.47}$	$\mathbf{49.58_{\pm 1.54}}$	$45.51_{\pm 1.54}$	$35.01_{\pm 4.40}$	$27.33_{\pm 2.81}$	$20.10_{\pm 0.86}$	$37.23_{\pm 12.64}$	$59.95_{\pm 17.46}$	$28.76_{\pm 2.94}$	$9.49_{\pm 2.33}$	$61.92_{\pm 1.29}$	$60.75_{\pm 6.24}$
10000.0	$39.96_{\pm 1.42}$	$49.52_{\pm 0.65}$	$\mathbf{47.67_{\pm 1.05}}$	$35.17_{\pm 2.25}$	$24.99_{\pm 1.90}$	$19.61_{\pm 0.42}$	$39.08_{\pm 10.89}$	$45.60_{\pm 2.86}$	$29.10_{\pm 2.17}$	$9.98_{\pm 1.27}$	$62.62_{\pm 2.07}$	$60.02_{\pm 7.27}$

Table 6: **Accuracy Statistical Table of MSFS on Shortcuts Maze Anti-Test.** The result of the optimal stretching ratio is marked in bold.

Model	Dataset F1 (%) Performance											
	Yelp					**Emotion**					**Beer**	
	Occur			**Style**		**Occur**			**Style**		**Concept**	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
0.001	$61.23_{\pm 0.57}$	$62.10_{\pm 0.41}$	$61.78_{\pm 0.33}$	$\mathbf{57.23}_{\pm \mathbf{0.71}}$	$57.98_{\pm 1.10}$	$70.55_{\pm 0.50}$	$75.78_{\pm 1.93}$	$79.43_{\pm 2.98}$	$\mathbf{60.90}_{\pm \mathbf{3.03}}$	$56.31_{\pm 3.59}$	$63.50_{\pm 0.84}$	$84.78_{\pm 0.96}$
0.01	$61.83_{\pm 0.29}$	$62.68_{\pm 0.12}$	$61.80_{\pm 1.26}$	$59.50_{\pm 0.57}$	$\mathbf{58.94}_{\pm \mathbf{0.21}}$	$74.63_{\pm 0.92}$	$73.93_{\pm 2.70}$	$81.52_{\pm 1.34}$	$68.61_{\pm 2.58}$	$\mathbf{59.38}_{\pm \mathbf{2.00}}$	$66.56_{\pm 1.43}$	$86.78_{\pm 0.44}$
0.1	$62.79_{\pm 1.14}$	$63.19_{\pm 0.08}$	$63.12_{\pm 0.55}$	$60.47_{\pm 0.60}$	$60.32_{\pm 0.35}$	$81.57_{\pm 2.08}$	$81.21_{\pm 0.55}$	$85.13_{\pm 1.31}$	$82.14_{\pm 1.12}$	$65.73_{\pm 1.71}$	$76.25_{\pm 0.69}$	$\mathbf{87.35}_{\pm \mathbf{0.95}}$
10.0	$63.95_{\pm 0.84}$	$63.64_{\pm 0.16}$	$62.86_{\pm 1.16}$	$59.41_{\pm 1.86}$	$59.60_{\pm 0.10}$	$83.20_{\pm 1.51}$	$82.58_{\pm 1.81}$	$82.26_{\pm 0.09}$	$78.15_{\pm 0.07}$	$59.56_{\pm 3.41}$	$75.68_{\pm 1.04}$	$88.32_{\pm 1.40}$
100.0	$\mathbf{63.48}_{\pm \mathbf{0.16}}$	$63.17_{\pm 0.14}$	$62.85_{\pm 0.50}$	$58.34_{\pm 0.87}$	$56.67_{\pm 0.22}$	$\mathbf{78.42}_{\pm \mathbf{1.74}}$	$\mathbf{73.57}_{\pm \mathbf{4.91}}$	$\mathbf{83.14}_{\pm \mathbf{0.41}}$	$75.90_{\pm 1.90}$	$56.26_{\pm 2.86}$	$\mathbf{73.20}_{\pm \mathbf{0.93}}$	$85.88_{\pm 1.06}$
1000.0	$63.35_{\pm 0.46}$	$\mathbf{62.08}_{\pm \mathbf{0.55}}$	$62.38_{\pm 1.02}$	$54.85_{\pm 3.12}$	$55.50_{\pm 0.87}$	$77.35_{\pm 1.47}$	$74.02_{\pm 0.92}$	$84.25_{\pm 1.12}$	$74.07_{\pm 3.14}$	$53.95_{\pm 0.91}$	$73.00_{\pm 0.69}$	$85.18_{\pm 1.51}$
10000.0	$60.94_{\pm 1.84}$	$62.21_{\pm 0.77}$	$\mathbf{61.05}_{\pm \mathbf{0.86}}$	$54.72_{\pm 1.31}$	$53.93_{\pm 0.68}$	$77.85_{\pm 2.24}$	$71.71_{\pm 2.52}$	$83.29_{\pm 1.47}$	$74.25_{\pm 1.64}$	$52.92_{\pm 0.28}$	$72.31_{\pm 1.41}$	$84.77_{\pm 2.01}$

Table 7: **F1 Statistical Table of MSFS on Shortcuts Maze Normal-Test.** The result of the optimal stretching ratio is marked in bold.

Model	Dataset F1 (%) Performance											
	Yelp					**Emotion**					**Beer**	
	Occur			**Style**		**Occur**			**Style**		**Concept**	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
0.001	$31.48_{\pm 0.55}$	$46.41_{\pm 0.22}$	$44.08_{\pm 1.27}$	$\mathbf{42.92}_{\pm \mathbf{0.74}}$	$27.30_{\pm 0.64}$	$28.61_{\pm 1.38}$	$44.20_{\pm 2.57}$	$41.19_{\pm 3.20}$	$\mathbf{28.31}_{\pm \mathbf{5.13}}$	$15.86_{\pm 1.69}$	$53.51_{\pm 4.32}$	$59.60_{\pm 3.61}$
0.01	$33.80_{\pm 1.49}$	$44.44_{\pm 2.09}$	$41.57_{\pm 1.71}$	$43.42_{\pm 0.41}$	$\mathbf{28.68}_{\pm \mathbf{0.73}}$	$28.48_{\pm 3.38}$	$43.88_{\pm 4.89}$	$46.81_{\pm 2.91}$	$32.29_{\pm 2.03}$	$\mathbf{17.26}_{\pm \mathbf{1.46}}$	$54.52_{\pm 2.69}$	$65.05_{\pm 5.21}$
0.1	$36.75_{\pm 1.94}$	$45.71_{\pm 1.41}$	$47.21_{\pm 1.25}$	$43.49_{\pm 0.47}$	$31.01_{\pm 1.33}$	$30.22_{\pm 2.20}$	$38.43_{\pm 2.42}$	$48.23_{\pm 5.02}$	$33.81_{\pm 1.28}$	$17.41_{\pm 0.95}$	$65.77_{\pm 0.35}$	$\mathbf{66.73}_{\pm \mathbf{2.68}}$
10.0	$36.71_{\pm 1.66}$	$44.73_{\pm 2.62}$	$41.47_{\pm 1.66}$	$33.27_{\pm 5.94}$	$29.47_{\pm 1.21}$	$33.74_{\pm 2.61}$	$42.12_{\pm 1.88}$	$52.87_{\pm 4.00}$	$29.96_{\pm 1.42}$	$16.27_{\pm 0.66}$	$63.78_{\pm 3.86}$	$62.85_{\pm 1.24}$
100.0	$\mathbf{42.42}_{\pm \mathbf{2.35}}$	$47.94_{\pm 1.70}$	$46.78_{\pm 1.60}$	$35.84_{\pm 1.79}$	$23.30_{\pm 3.27}$	$\mathbf{32.38}_{\pm \mathbf{2.21}}$	$\mathbf{44.00}_{\pm \mathbf{5.28}}$	$\mathbf{65.16}_{\pm \mathbf{1.75}}$	$26.47_{\pm 3.17}$	$14.14_{\pm 0.87}$	$\mathbf{62.43}_{\pm \mathbf{1.51}}$	$61.92_{\pm 5.03}$
1000.0	$37.62_{\pm 2.17}$	$\mathbf{49.64}_{\pm \mathbf{1.71}}$	$44.53_{\pm 2.18}$	$31.92_{\pm 5.01}$	$24.23_{\pm 3.82}$	$28.13_{\pm 1.39}$	$38.75_{\pm 6.45}$	$57.97_{\pm 9.77}$	$26.22_{\pm 1.66}$	$10.64_{\pm 3.04}$	$60.46_{\pm 1.24}$	$62.11_{\pm 5.39}$
10000.0	$37.71_{\pm 2.21}$	$49.82_{\pm 0.92}$	$\mathbf{47.03}_{\pm \mathbf{1.17}}$	$32.56_{\pm 2.63}$	$21.12_{\pm 2.01}$	$28.58_{\pm 1.94}$	$40.97_{\pm 6.86}$	$49.65_{\pm 1.14}$	$28.86_{\pm 4.10}$	$10.65_{\pm 1.52}$	$60.74_{\pm 1.97}$	$61.43_{\pm 6.50}$

Table 8: **F1 Statistical Table of MSFS on Shortcuts Maze Anti-Test.** The result of the optimal stretching ratio is marked in bold.

Model	Dataset Accuracy (%) Performance											
	Yelp					**Emotion**					**Beer**	
	Occur			**Style**		**Occur**			**Style**		**Concept**	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
0.001	$\mathbf{62.43}_{\pm \mathbf{0.22}}$	$61.60_{\pm 0.19}$	$61.94_{\pm 0.60}$	$55.77_{\pm 0.79}$	$57.57_{\pm 0.54}$	$66.03_{\pm 5.82}$	$73.67_{\pm 6.56}$	$80.97_{\pm 3.25}$	~~$33.63_{\pm 10.56}$~~	$66.42_{\pm 3.16}$	$53.25_{\pm 2.34}$	$78.97_{\pm 1.20}$
0.01	$61.61_{\pm 0.16}$	$61.98_{\pm 0.39}$	$61.91_{\pm 0.18}$	$55.55_{\pm 0.47}$	$56.97_{\pm 0.52}$	$\mathbf{63.94}_{\pm \mathbf{9.37}}$	$72.60_{\pm 7.87}$	$83.02_{\pm 2.67}$	~~$30.71_{\pm 6.17}$~~	$65.64_{\pm 5.62}$	$\mathbf{51.25}_{\pm \mathbf{5.01}}$	$79.47_{\pm 1.31}$
0.1	$61.85_{\pm 0.30}$	$\mathbf{62.20}_{\pm \mathbf{0.34}}$	$61.46_{\pm 0.35}$	$54.93_{\pm 0.81}$	$57.12_{\pm 0.56}$	$76.01_{\pm 0.79}$	$75.28_{\pm 6.63}$	$83.60_{\pm 1.99}$	~~$32.94_{\pm 9.54}$~~	$67.74_{\pm 1.78}$	$52.52_{\pm 3.32}$	$80.78_{\pm 0.52}$
10.0	$61.83_{\pm 0.37}$	$62.04_{\pm 0.22}$	$61.53_{\pm 0.54}$	$55.81_{\pm 0.25}$	$57.66_{\pm 0.33}$	$71.63_{\pm 0.89}$	$71.58_{\pm 6.34}$	$80.78_{\pm 2.11}$	~~$36.40_{\pm 7.03}$~~	$67.49_{\pm 5.80}$	$50.55_{\pm 3.13}$	$\mathbf{80.68}_{\pm \mathbf{1.39}}$
100.0	$62.05_{\pm 0.39}$	$61.75_{\pm 0.42}$	$61.49_{\pm 0.24}$	$\mathbf{54.79}_{\pm \mathbf{0.59}}$	$\mathbf{57.05}_{\pm \mathbf{0.25}}$	$70.41_{\pm 4.71}$	$\mathbf{69.73}_{\pm \mathbf{5.80}}$	$82.87_{\pm 2.11}$	~~$31.14_{\pm 6.57}$~~	$66.76_{\pm 1.39}$	$51.75_{\pm 1.88}$	$80.13_{\pm 1.47}$
1000.0	$61.77_{\pm 0.46}$	$61.76_{\pm 0.06}$	$61.42_{\pm 0.47}$	$56.35_{\pm 0.55}$	$57.30_{\pm 0.48}$	$71.48_{\pm 2.09}$	$73.53_{\pm 5.28}$	$\mathbf{80.19}_{\pm \mathbf{2.46}}$	~~$30.95_{\pm 4.03}$~~	$67.15_{\pm 2.79}$	$51.27_{\pm 2.73}$	$80.90_{\pm 1.74}$
10000.0	$62.44_{\pm 0.19}$	$61.75_{\pm 0.36}$	$\mathbf{61.39}_{\pm \mathbf{0.45}}$	$55.41_{\pm 0.99}$	$56.35_{\pm 0.62}$	$66.72_{\pm 3.10}$	$75.72_{\pm 7.00}$	$83.07_{\pm 0.21}$	~~$29.68_{\pm 8.24}$~~	$\mathbf{63.55}_{\pm \mathbf{4.98}}$	$51.02_{\pm 3.83}$	$79.77_{\pm 1.54}$

Table 9: **Accuracy Statistical Table of MSFS*Ds* on Shortcuts Maze Normal-Test.** The result of the optimal stretching ratio is marked in bold.

Model	Dataset Accuracy (%) Performance											
	Yelp					**Emotion**					**Beer**	
	Occur			**Style**		**Occur**			**Style**		**Concept**	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
0.001	$\mathbf{39.84}_{\pm \mathbf{2.14}}$	$47.21_{\pm 0.95}$	$49.32_{\pm 2.35}$	$44.65_{\pm 0.65}$	$35.77_{\pm 0.57}$	$16.55_{\pm 4.25}$	$36.16_{\pm 11.03}$	$45.99_{\pm 14.10}$	~~$15.18_{\pm 2.33}$~~	$15.09_{\pm 0.73}$	$40.32_{\pm 0.76}$	$61.08_{\pm 5.26}$
0.01	$37.96_{\pm 1.32}$	$46.97_{\pm 1.55}$	$50.25_{\pm 1.59}$	$45.13_{\pm 0.78}$	$36.20_{\pm 0.27}$	$\mathbf{19.56}_{\pm \mathbf{6.02}}$	$35.04_{\pm 11.35}$	$38.54_{\pm 5.24}$	~~$14.45_{\pm 1.67}$~~	$15.33_{\pm 0.63}$	$\mathbf{42.13}_{\pm \mathbf{4.29}}$	$55.07_{\pm 3.81}$
0.1	$37.43_{\pm 1.99}$	$\mathbf{48.52}_{\pm \mathbf{2.24}}$	$51.57_{\pm 0.88}$	$45.01_{\pm 0.64}$	$36.01_{\pm 0.41}$	$14.50_{\pm 2.16}$	$32.77_{\pm 11.75}$	$38.54_{\pm 2.93}$	~~$14.70_{\pm 1.61}$~~	$14.94_{\pm 0.18}$	$37.15_{\pm 2.63}$	$57.30_{\pm 2.32}$
10.0	$37.73_{\pm 0.81}$	$47.73_{\pm 0.68}$	$50.30_{\pm 2.50}$	$45.33_{\pm 0.48}$	$35.37_{\pm 0.18}$	$16.30_{\pm 2.13}$	$35.96_{\pm 8.23}$	$39.56_{\pm 14.28}$	~~$14.94_{\pm 2.21}$~~	$15.04_{\pm 0.93}$	$38.83_{\pm 5.57}$	$\mathbf{63.68}_{\pm \mathbf{3.47}}$
100.0	$37.36_{\pm 1.30}$	$47.55_{\pm 1.87}$	$48.92_{\pm 3.56}$	$\mathbf{45.03}_{\pm \mathbf{0.57}}$	$\mathbf{36.44}_{\pm \mathbf{0.45}}$	$14.45_{\pm 2.30}$	$\mathbf{38.00}_{\pm \mathbf{10.13}}$	$43.36_{\pm 7.82}$	~~$14.94_{\pm 1.74}$~~	$15.47_{\pm 0.73}$	$39.35_{\pm 6.00}$	$56.78_{\pm 2.52}$
1000.0	$37.76_{\pm 0.44}$	$47.01_{\pm 1.27}$	$50.10_{\pm 2.59}$	$45.07_{\pm 0.64}$	$36.10_{\pm 0.48}$	$18.54_{\pm 4.88}$	$33.82_{\pm 9.72}$	$\mathbf{51.34}_{\pm \mathbf{1.03}}$	~~$14.94_{\pm 1.44}$~~	$15.33_{\pm 0.86}$	$37.03_{\pm 5.75}$	$59.38_{\pm 2.13}$
10000.0	$39.29_{\pm 0.43}$	$47.17_{\pm 2.46}$	$\mathbf{51.96}_{\pm \mathbf{0.95}}$	$45.12_{\pm 0.28}$	$35.56_{\pm 0.80}$	$20.83_{\pm 7.87}$	$38.20_{\pm 14.75}$	$38.73_{\pm 5.06}$	~~$13.72_{\pm 1.14}$~~	$\mathbf{15.86}_{\pm \mathbf{1.31}}$	$37.60_{\pm 1.08}$	$60.35_{\pm 7.84}$

Table 10: **Accuracy Statistical Table of MSFS*Ds* on Shortcuts Maze Anti-Test.** The result of the optimal stretching ratio is marked in bold.

Model	Dataset F1 (%) Performance											
	Yelp					Emotion					Beer	
	Occur			Style		Occur			Style		Concept	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
0.001	**62.07**$_{\pm 0.04}$	61.28$_{\pm 0.29}$	61.70$_{\pm 0.45}$	55.70$_{\pm 0.70}$	57.57$_{\pm 0.32}$	52.49$_{\pm 2.34}$	59.86$_{\pm 5.33}$	70.09$_{\pm 3.43}$	~~32.40$_{\pm 6.22}$~~	56.46$_{\pm 0.40}$	52.39$_{\pm 3.39}$	78.96$_{\pm 0.98}$
0.01	60.66$_{\pm 0.49}$	61.51$_{\pm 0.59}$	61.21$_{\pm 0.58}$	55.84$_{\pm 0.52}$	56.93$_{\pm 0.34}$	**52.21**$_{\pm 5.41}$	58.76$_{\pm 6.06}$	70.84$_{\pm 3.42}$	~~29.25$_{\pm 4.48}$~~	57.90$_{\pm 2.79}$	**51.11**$_{\pm 5.38}$	79.57$_{\pm 1.24}$
0.1	60.99$_{\pm 0.36}$	**61.82**$_{\pm 0.48}$	61.03$_{\pm 0.49}$	54.79$_{\pm 1.00}$	56.90$_{\pm 0.95}$	59.44$_{\pm 2.51}$	61.69$_{\pm 4.61}$	72.55$_{\pm 2.54}$	~~31.30$_{\pm 6.28}$~~	57.44$_{\pm 1.26}$	51.69$_{\pm 3.87}$	80.76$_{\pm 0.53}$
10.0	61.05$_{\pm 0.39}$	61.90$_{\pm 0.48}$	60.82$_{\pm 0.44}$	55.67$_{\pm 0.45}$	57.56$_{\pm 0.17}$	56.72$_{\pm 0.37}$	58.28$_{\pm 4.62}$	68.90$_{\pm 1.50}$	~~33.74$_{\pm 4.48}$~~	56.81$_{\pm 4.40}$	50.04$_{\pm 3.77}$	**80.50**$_{\pm 1.26}$
100.0	61.17$_{\pm 0.65}$	60.89$_{\pm 0.35}$	60.42$_{\pm 0.28}$	**54.30**$_{\pm 1.10}$	**57.17**$_{\pm 0.45}$	56.11$_{\pm 3.53}$	**56.43**$_{\pm 4.73}$	70.52$_{\pm 3.37}$	~~29.92$_{\pm 4.39}$~~	57.39$_{\pm 1.78}$	50.71$_{\pm 1.58}$	80.11$_{\pm 1.27}$
1000.0	60.90$_{\pm 0.60}$	61.09$_{\pm 0.22}$	60.67$_{\pm 0.45}$	56.48$_{\pm 0.53}$	57.39$_{\pm 0.25}$	56.53$_{\pm 0.76}$	59.37$_{\pm 3.99}$	**68.21**$_{\pm 3.06}$	~~29.17$_{\pm 3.57}$~~	55.72$_{\pm 5.05}$	50.98$_{\pm 2.77}$	80.87$_{\pm 1.66}$
10000.0	61.57$_{\pm 0.24}$	61.33$_{\pm 0.39}$	**60.48**$_{\pm 0.67}$	55.47$_{\pm 0.92}$	55.43$_{\pm 0.58}$	53.36$_{\pm 1.80}$	62.05$_{\pm 4.79}$	72.02$_{\pm 0.51}$	~~28.63$_{\pm 3.73}$~~	**56.49**$_{\pm 3.09}$	51.02$_{\pm 4.46}$	79.86$_{\pm 1.37}$

Table 11: **F1 Statistical Table of MSFS*Ds* on Shortcuts Maze Normal-Test.** The result of the optimal stretching ratio is marked in bold.

Model	Dataset F1 (%) Performance											
	Yelp					Emotion					Beer	
	Occur			Style		Occur			Style		Concept	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
0.001	**38.03**$_{\pm 3.37}$	47.33$_{\pm 1.23}$	49.72$_{\pm 2.47}$	41.95$_{\pm 0.56}$	27.93$_{\pm 0.58}$	23.92$_{\pm 1.46}$	37.60$_{\pm 4.79}$	45.17$_{\pm 4.85}$	~~16.26$_{\pm 0.68}$~~	14.38$_{\pm 0.67}$	40.40$_{\pm 0.74}$	62.89$_{\pm 5.03}$
0.01	36.36$_{\pm 2.39}$	46.98$_{\pm 1.82}$	50.50$_{\pm 1.75}$	42.33$_{\pm 0.80}$	28.46$_{\pm 0.46}$	**26.60**$_{\pm 1.63}$	36.84$_{\pm 5.76}$	41.44$_{\pm 2.15}$	~~16.49$_{\pm 0.84}$~~	14.61$_{\pm 1.14}$	**42.28**$_{\pm 4.73}$	57.39$_{\pm 3.39}$
0.1	35.22$_{\pm 2.44}$	**48.76**$_{\pm 2.51}$	52.05$_{\pm 0.89}$	42.16$_{\pm 0.67}$	28.25$_{\pm 0.48}$	21.55$_{\pm 1.61}$	36.99$_{\pm 7.39}$	40.32$_{\pm 2.08}$	~~16.80$_{\pm 0.64}$~~	14.63$_{\pm 0.75}$	37.09$_{\pm 3.02}$	59.07$_{\pm 2.22}$
10.0	35.35$_{\pm 1.01}$	47.85$_{\pm 0.60}$	50.69$_{\pm 2.67}$	42.75$_{\pm 0.41}$	27.98$_{\pm 0.54}$	24.35$_{\pm 1.17}$	36.61$_{\pm 4.96}$	42.61$_{\pm 6.62}$	~~16.50$_{\pm 0.79}$~~	14.16$_{\pm 0.84}$	38.83$_{\pm 6.19}$	**64.97**$_{\pm 3.24}$
100.0	35.22$_{\pm 2.36}$	47.68$_{\pm 2.04}$	49.09$_{\pm 4.07}$	**41.95**$_{\pm 0.56}$	**28.43**$_{\pm 0.47}$	22.28$_{\pm 1.89}$	**38.11**$_{\pm 5.07}$	42.81$_{\pm 5.21}$	~~17.18$_{\pm 0.75}$~~	14.96$_{\pm 0.86}$	39.61$_{\pm 6.29}$	58.83$_{\pm 2.73}$
1000.0	35.18$_{\pm 0.82}$	47.19$_{\pm 1.47}$	50.22$_{\pm 3.10}$	42.22$_{\pm 0.42}$	28.29$_{\pm 0.57}$	26.73$_{\pm 2.53}$	36.12$_{\pm 5.53}$	**47.58**$_{\pm 0.47}$	~~16.43$_{\pm 0.97}$~~	14.42$_{\pm 1.56}$	36.34$_{\pm 6.60}$	61.38$_{\pm 2.32}$
10000.0	37.67$_{\pm 0.66}$	47.19$_{\pm 2.81}$	**52.27**$_{\pm 0.81}$	42.09$_{\pm 0.14}$	27.63$_{\pm 0.44}$	27.17$_{\pm 4.51}$	39.54$_{\pm 8.66}$	42.50$_{\pm 1.34}$	~~16.19$_{\pm 1.01}$~~	**15.40**$_{\pm 0.84}$	36.87$_{\pm 2.14}$	62.23$_{\pm 7.27}$

Table 12: **F1 Statistical Table of MSFS*Ds* on Shortcuts Maze Anti-Test.** The result of the optimal stretching ratio is marked in bold.

Model	Dataset △ (%) Performance											
	Yelp					Emotion					Beer	
	Occur			Style		Occur			Style		Concept	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
DeEBRT	24.09	20.66	25.31	16.52	24.90	59.56	51.53	**3.79**	58.54	60.00	**14.85**	34.95
+MSFS	**21.53**	**6.54**	**5.60**	**10.32**	**17.94**	**49.19**	**24.37**	7.73	**47.15**	**55.03**	18.14	**21.34**
RoBERTa	26.49	14.96	21.94	18.27	23.52	69.92	55.18	35.32	58.54	55.32	16.84	22.19
+MSFS	**17.69**	**-5.81**	**7.46**	**7.97**	**22.41**	**54.74**	**29.34**	**0.14**	**42.48**	**40.72**	**15.40**	**20.65**
GPT2	22.18	9.22	13.54	18.10	25.85	65.40	44.81	7.59	**54.89**	57.51	14.20	8.30
+MSFS	**14.52**	**5.83**	**0.49**	**13.07**	**15.00**	**41.31**	**27.59**	**3.79**	-	**42.62**	**7.70**	**5.89**
GPT2-Medium	20.81	17.97	24.55	17.58	26.72	68.46	31.09	64.52	62.77	57.51	16.95	11.65
+MSFS	**17.59**	**7.09**	**7.56**	**15.18**	**22.68**	**60.43**	**8.32**	**1.60**	**1.31**	**52.26**	**6.65**	**-1.94**

Table 13: **Statistical Table of *△* Performance Results of MSFS with Different Backbone Models on the Shortcuts Maze Benchmark.** The test set accuracy performance of single backbone models is provided in Appendix Table 14 – 21.

Model	**Dataset △ (%) Performance**											
	Yelp					**Emotion**					**Beer**	
	Occur			**Style**		**Occur**			**Style**		**Concept**	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
ERM	67.88	67.59	67.55	64.50	62.92	82.92	91.97	91.97	86.86	73.28	79.00	92.45
0.001	52.87	53.38	45.57	**46.76**	**49.74**	73.14	65.99	65.55	~~50.66~~	~~9.20~~	59.00	~~56.30~~
0.01	53.58	~~20.00~~	52.74	51.16	51.92	70.95	73.43	69.64	~~10.66~~	~~15.47~~	**51.35**	~~59.65~~
0.1	57.13	56.01	55.26	55.16	55.26	72.85	79.42	70.36	~~10.36~~	**68.91**	57.10	60.75
10.0	58.25	55.83	57.56	56.02	51.80	79.71	78.10	**81.31**	81.61	67.74	62.30	72.20
100.0	58.42	56.97	56.35	56.20	54.40	**79.42**	74.45	76.93	78.83	72.26	58.25	67.50
1000.0	58.04	**55.25**	**55.50**	56.32	49.08	76.79	77.08	78.25	82.34	70.80	62.30	**70.70**
10000.0	**57.89**	56.40	54.73	56.16	54.38	74.89	**78.83**	75.77	**75.91**	68.47	62.00	68.20

Table 14: **Statistical Results of Accuracy of the MSFS Method Based on DeBERT on the Normal-Test Set.**

Model	**Dataset △ (%) Performance**											
	Yelp					**Emotion**					**Beer**	
	Occur			**Style**		**Occur**			**Style**		**Concept**	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
ERM	43.78	46.92	42.24	47.98	38.02	23.36	40.44	88.18	28.32	13.28	64.15	57.50
0.001	31.22	36.68	20.00	**36.44**	**31.80**	15.62	21.17	43.80	~~17.66~~	~~9.20~~	37.85	~~38.45~~
0.01	23.47	~~36.21~~	44.07	34.76	26.66	16.35	28.32	37.37	~~9.20~~	~~10.51~~	**33.20**	~~36.45~~
0.1	31.83	43.07	41.63	37.22	28.24	18.98	29.64	46.57	~~9.20~~	**13.87**	28.10	39.30
10.0	35.27	39.00	42.52	35.68	20.02	22.34	38.69	**73.58**	31.97	5.69	39.35	44.90
100.0	36.47	50.15	42.52	34.52	17.88	**30.22**	44.23	66.13	27.59	7.01	36.00	41.80
1000.0	36.50	**48.71**	**49.90**	34.04	17.54	26.42	44.38	69.20	32.55	7.01	39.85	**49.35**
10000.0	**36.36**	44.24	46.72	35.92	21.30	23.94	**54.45**	67.74	**28.76**	5.11	39.60	41.75

Table 15: **Statistical Results of Accuracy of the MSFS Method Based on DeBERT on the Anti-Test Set.**

Model	Dataset △ (%) Performance											
	Yelp					**Emotion**					**Beer**	
	Occur			**Style**		**Occur**			**Style**		**Concept**	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
ERM	65.45	65.16	65.74	61.42	60.34	90.36	90.51	89.78	86.13	70.36	74.70	89.55
0.001	65.26	64.80	63.69	**58.10**	**57.00**	85.55	85.99	84.96	~~20.88~~	65.26	52.75	**84.25**
0.01	64.72	64.72	**62.42**	60.52	57.62	85.99	83.94	89.64	**62.04**	**61.17**	63.70	86.50
0.1	64.77	65.49	64.49	60.86	60.66	89.93	90.22	87.15	85.84	75.18	70.50	88.20
10.0	65.79	65.15	64.95	61.74	60.10	88.18	**91.68**	88.76	84.53	71.24	76.35	90.45
100.0	**64.09**	**47.64**	64.01	60.10	56.92	87.01	87.30	**85.40**	85.99	72.12	71.40	82.25
1000.0	64.30	63.42	63.91	59.84	57.98	**85.69**	87.59	89.05	89.34	70.22	66.90	72.25
10000.0	63.85	64.46	63.75	58.84	57.34	85.40	89.20	91.39	84.67	71.24	**67.15**	82.20

Table 16: **Statistical Results of Accuracy of the MSFS Method Based on RoBERT on the Normal-Test Set.**

Model	Dataset △ (%) Performance											
	Yelp					**Emotion**					**Beer**	
	Occur			**Style**		**Occur**			**Style**		**Concept**	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
ERM	38.95	50.19	43.79	43.14	36.82	20.44	35.33	54.45	27.59	15.04	57.85	67.35
0.001	42.31	47.62	53.75	**50.12**	**34.58**	20.73	43.65	28.03	~~21.02~~	19.85	27.10	**63.60**
0.01	41.61	54.32	**54.96**	47.78	34.82	16.35	31.39	30.66	**19.56**	**20.44**	38.00	56.60
0.1	40.54	49.85	51.89	46.56	35.08	20.00	34.60	27.30	28.76	15.04	51.40	53.85
10.0	43.61	52.52	51.76	40.80	26.10	18.25	**62.34**	32.41	28.03	13.43	60.10	58.85
100.0	**46.40**	**53.46**	47.17	35.26	22.34	25.99	57.81	**85.26**	34.60	11.82	53.00	48.60
1000.0	45.37	52.46	53.83	37.38	19.84	**30.95**	56.64	34.01	31.97	11.24	48.45	50.60
10000.0	43.96	51.40	54.47	32.04	27.18	20.44	57.81	35.91	26.42	9.78	**51.75**	45.65

Table 17: **Statistical Results of Accuracy of the MSFS Method Based on RoBERT on the Anti-Test Set.**

Model	Dataset △ (%) Performance											
	Yelp					**Emotion**					**Beer**	
	Occur			**Style**		**Occur**			**Style**		**Concept**	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
ERM	64.68	63.26	64.95	61.28	59.92	84.67	80.44	89.05	79.71	71.09	69.85	85.55
0.001	64.31	62.56	63.99	**50.84**	**49.60**	**57.81**	63.50	77.96	~~28.18~~	63.36	66.80	77.65
0.01	63.92	63.06	63.46	55.32	54.74	62.19	64.53	79.27	~~24.53~~	64.82	69.00	78.10
0.1	63.27	63.09	64.25	60.28	60.16	67.59	75.33	84.67	~~21.02~~	74.01	69.70	**78.85**
10.0	63.26	63.26	62.72	59.08	58.98	71.97	66.57	**79.27**	~~29.93~~	**52.41**	**65.70**	70.15
100.0	59.84	60.02	59.81	56.34	55.82	66.57	61.31	80.88	~~23.94~~	61.75	57.10	58.55
1000.0	59.83	**58.13**	**58.98**	54.52	56.38	66.72	**61.75**	79.85	~~28.91~~	63.36	56.30	60.60
10000.0	**58.30**	59.65	58.37	53.68	55.96	70.51	69.78	75.33	~~21.90~~	56.93	57.05	60.10

Table 18: **Statistical Results of Accuracy of the MSFS Method Based on GPT2 on the Normal-Test Set.**

Model	Dataset △ (%) Performance											
	Yelp					**Emotion**					**Beer**	
	Occur			**Style**		**Occur**			**Style**		**Concept**	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
ERM	42.50	54.04	51.40	43.18	34.06	19.27	35.62	81.46	24.82	13.58	55.65	77.25
0.001	42.06	51.14	54.79	**37.76**	**34.60**	**16.50**	28.32	65.40	~~25.26~~	12.26	49.85	69.80
0.01	41.63	50.92	56.31	37.42	31.78	16.20	29.64	67.01	~~19.71~~	12.26	50.45	71.70
0.1	44.16	53.96	55.44	42.02	32.14	14.74	32.99	80.29	~~35.04~~	11.68	54.95	**72.95**
10.0	44.74	52.77	56.50	37.52	32.80	19.27	34.74	**75.47**	~~33.28~~	**9.78**	**58.00**	58.40
100.0	44.70	52.21	58.89	35.96	31.42	20.58	30.80	69.64	~~18.83~~	10.22	47.95	43.40
1000.0	44.90	**52.30**	**58.49**	31.94	31.22	19.42	**34.16**	66.86	~~16.79~~	9.78	46.20	47.60
10000.0	**43.77**	51.54	57.30	36.90	31.10	22.34	38.10	68.47	~~25.11~~	9.34	42.90	43.70

Table 19: **Statistical Results of Accuracy of the MSFS Method Based on GPT2 on the Anti-Test Set.**

Model	Dataset △ (%) Performance											
	Yelp					**Emotion**					**Beer**	
	Occur			**Style**		**Occur**			**Style**		**Concept**	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
ERM	66.92	66.78	66.73	63.90	61.68	92.85	91.09	91.82	90.22	77.23	80.40	93.65
0.001	66.02	65.12	65.91	**62.20**	60.80	85.99	87.59	90.51	80.88	69.49	77.20	90.15
0.01	65.54	65.45	65.55	62.82	60.94	88.03	85.40	90.80	77.37	68.91	77.50	91.50
0.1	66.51	65.31	66.52	63.04	63.28	90.22	82.34	91.24	73.14	74.45	79.40	91.35
10.0	**66.64**	66.59	67.35	63.28	61.14	88.32	87.74	92.26	87.30	75.91	**79.65**	90.50
100.0	66.27	66.40	67.12	61.66	61.10	**81.31**	80.44	89.78	83.21	74.31	76.20	89.20
1000.0	66.63	**65.15**	**65.81**	61.22	59.76	84.67	**81.75**	89.49	84.67	**72.85**	75.60	89.10
10000.0	65.94	66.25	66.13	60.34	**58.88**	83.65	81.90	**88.18**	**83.36**	73.14	75.25	**83.85**

Table 20: **Statistical Results of Accuracy of the MSFS Method Based on GPT2-medium on the Normal-Test Set.**

Model	Dataset △ (%) Performance											
	Yelp					**Emotion**					**Beer**	
	Occur			**Style**		**Occur**			**Style**		**Concept**	
	ST	Syn	Catg	Reg	Auth	ST	Syn	Catg	Reg	Auth	Occur	Corr
ERM	46.11	48.80	42.17	46.32	34.96	24.38	60.00	27.30	27.45	19.71	63.45	82.00
0.001	44.99	55.65	48.91	**47.02**	35.72	21.17	71.24	43.21	69.78	12.55	67.90	84.55
0.01	47.68	51.24	50.43	45.40	35.16	20.29	67.74	53.58	81.31	10.66	68.15	81.65
0.1	45.83	51.58	48.67	46.00	36.26	21.17	69.05	71.97	81.17	13.58	71.35	84.25
10.0	**49.05**	52.97	49.56	46.74	35.98	20.73	65.69	69.93	79.56	18.98	**73.00**	86.05
100.0	43.61	53.89	55.63	40.42	34.68	**20.88**	70.07	85.69	80.15	12.70	67.00	86.05
1000.0	46.61	**58.05**	**58.24**	42.00	35.90	19.85	**73.43**	85.40	80.44	**20.58**	66.30	83.90
10000.0	48.28	56.33	55.89	41.08	**36.20**	17.66	71.24	**86.57**	**82.04**	18.39	67.15	**85.80**

Table 21: **Statistical Results of Accuracy of the MSFS Method Based on GPT2-medium on the Anti-Test Set.**

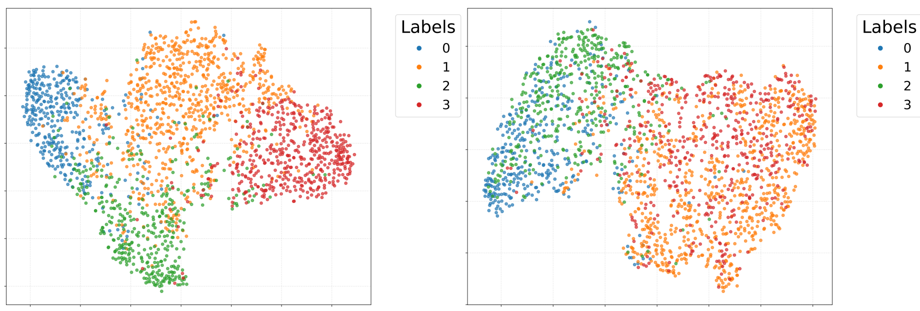

Fig. 1: **Visualization analysis graph of the test set embedding projection points at a stretching ratio of 0.001.** The left panel shows the embedding projection points of the normal-test set, and the right panel shows those of the anti-test set. The Δ for this result is 29.90%, where the accuracy result of the normal-test set is 89.40% and that of the anti-test set is 59.50%.

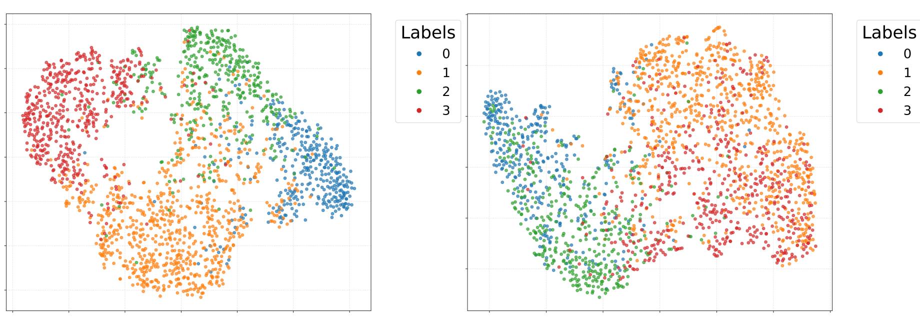

Fig. 2: **Visualization analysis graph of the test set embedding projection points at a stretching ratio of 0.01.** The left panel shows the embedding projection points of the normal-test set, and the right panel shows those of the anti-test set. The Δ for this result is 24.90%, where the accuracy result of the normal-test set is 86.05% and that of the anti-test set is 61.15%.

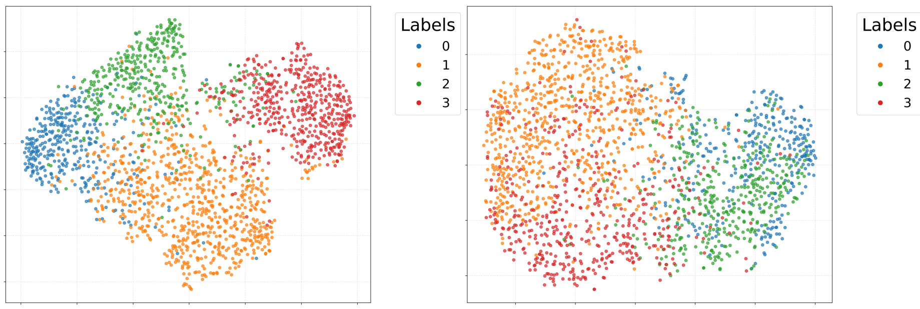

Fig. 3: **Visualization analysis graph of the test set embedding projection points at a stretching ratio of 10.** The left panel shows the embedding projection points of the normal-test set, and the right panel shows those of the anti-test set. The Δ for this result is 29.50%, where the accuracy result of the normal-test set is 89.55% and that of the anti-test set is 60.05%.

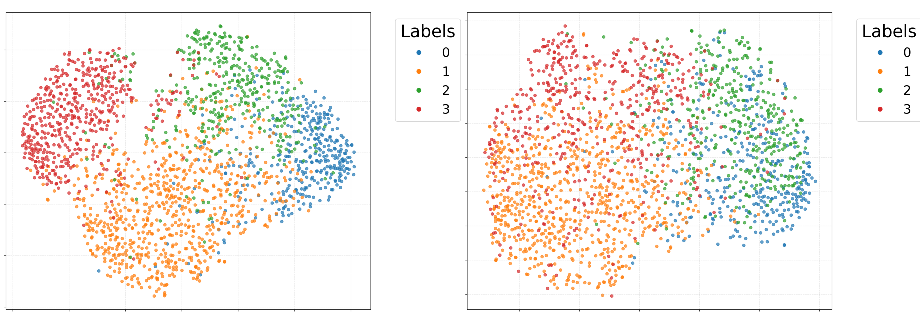

Fig. 4: **Visualization analysis graph of the test set embedding projection points at a stretching ratio of 100.** The left panel shows the embedding projection points of the normal-test set, and the right panel shows those of the anti-test set. The Δ for this result is 23.45%, where the accuracy result of the normal-test set is 86.50% and that of the anti-test set is 63.05%.

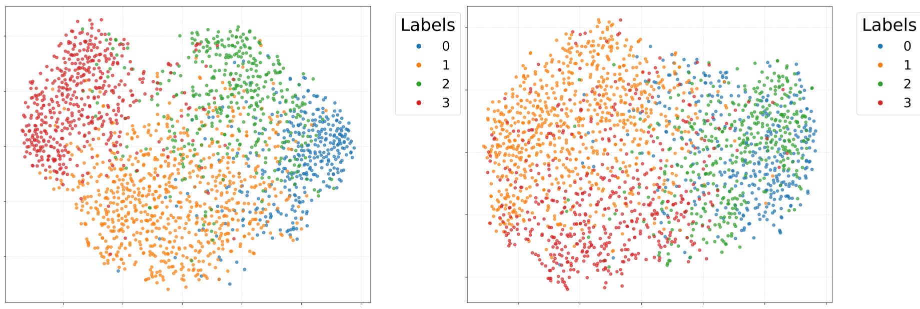

Fig. 5: **Visualization analysis graph of the test set embedding projection points at a stretching ratio of 1000.** The left panel shows the embedding projection points of the normal-test set, and the right panel shows those of the anti-test set. The Δ for this result is 24.75%, where the accuracy result of the normal-test set is 86.50% and that of the anti-test set is 61.75%.

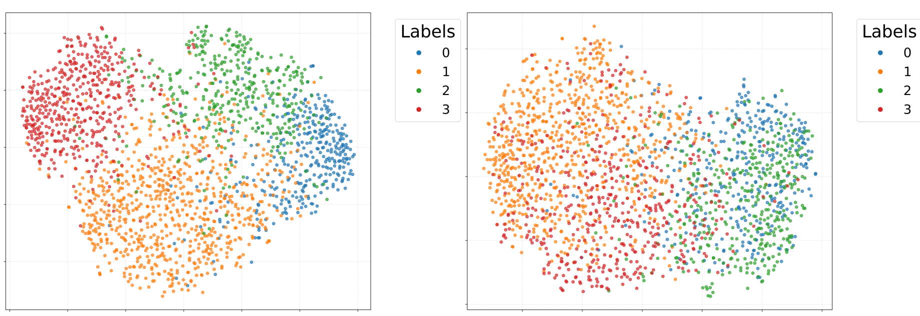

Fig. 6: **Visualization analysis graph of the test set embedding projection points at a stretching ratio of 10000.** The left panel shows the embedding projection points of the normal-test set, and the right panel shows those of the anti-test set. The Δ for this result is 35.90%, where the accuracy result of the normal-test set is 87.35% and that of the anti-test set is 51.45%.

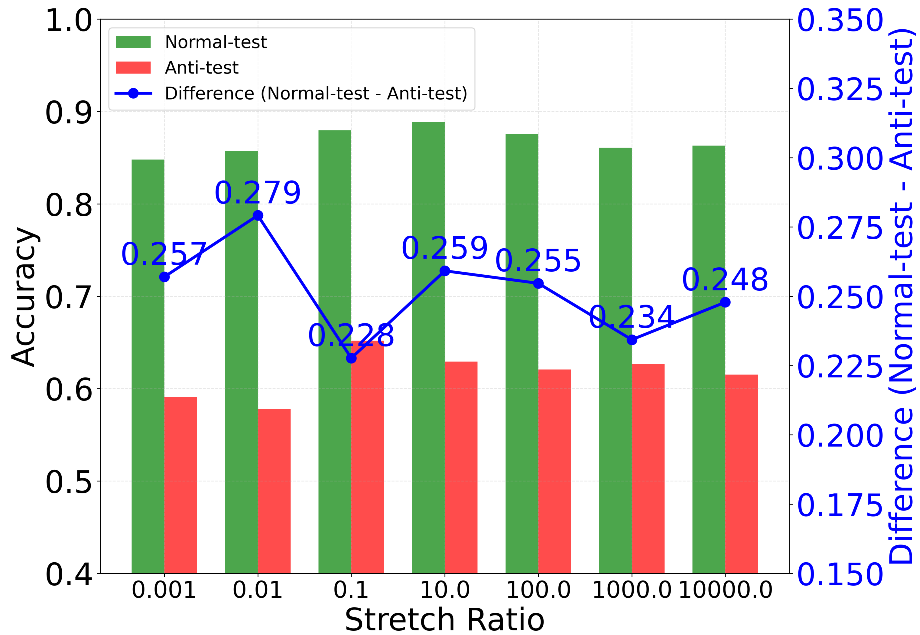

Fig. 7: **Experimental Results Figure with the average embedding of whole samples as the Stretching Center**

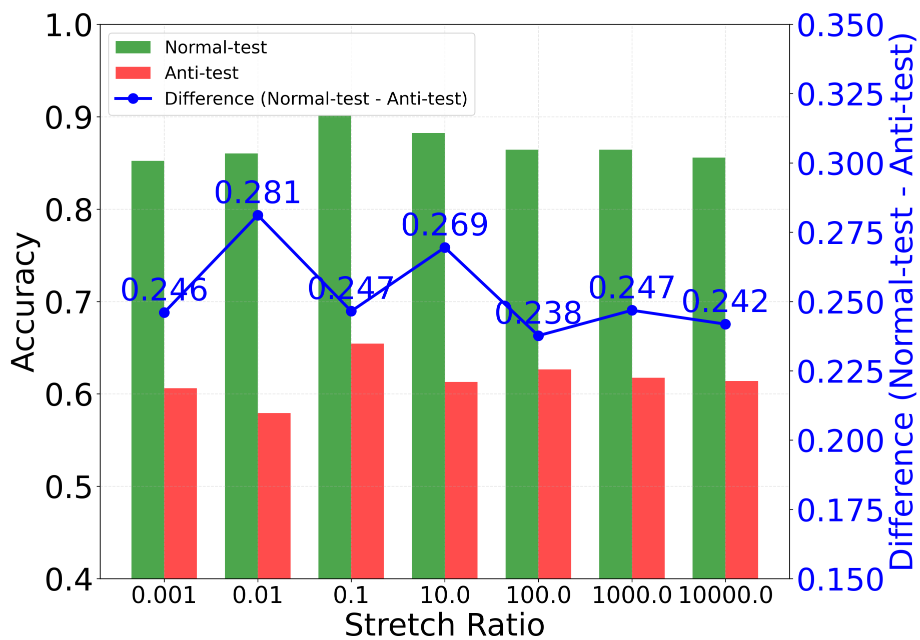

Fig. 8: **Experimental Results Figure with random embedding as the Stretching Center**

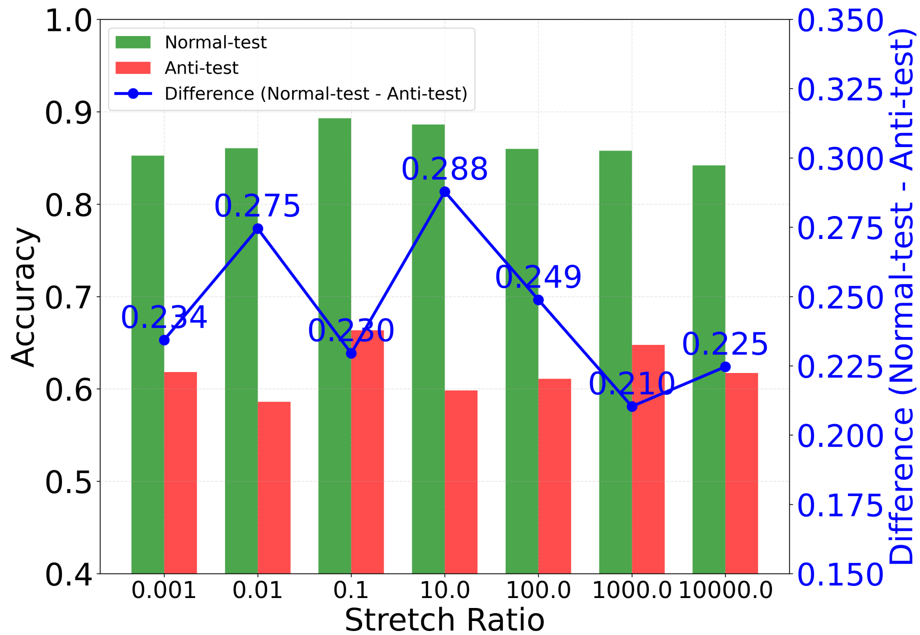

Fig. 9: **Experimental Results Figure with the random text 1 embedding as the Stretching Center**

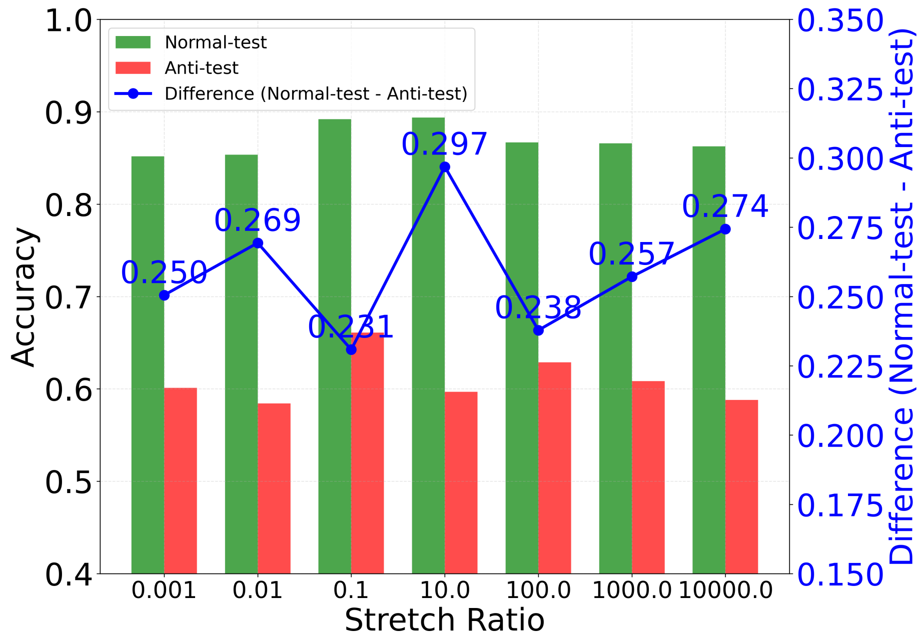

Fig. 10: **Experimental Results Figure with the random text 2 embedding as the Stretching Center**

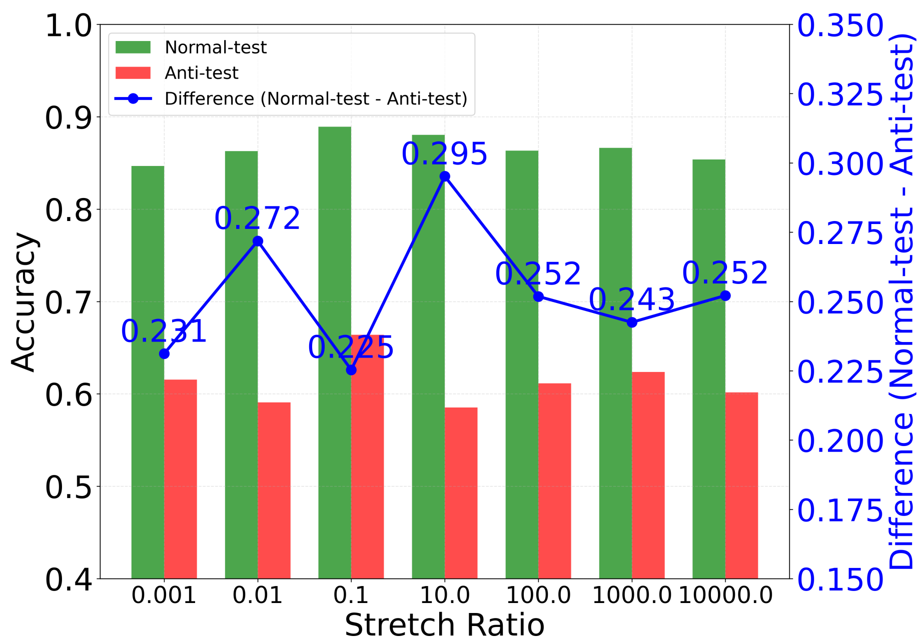

Fig. 11: **Experimental Results Figure with the random text 3 embedding as the Stretching Center**

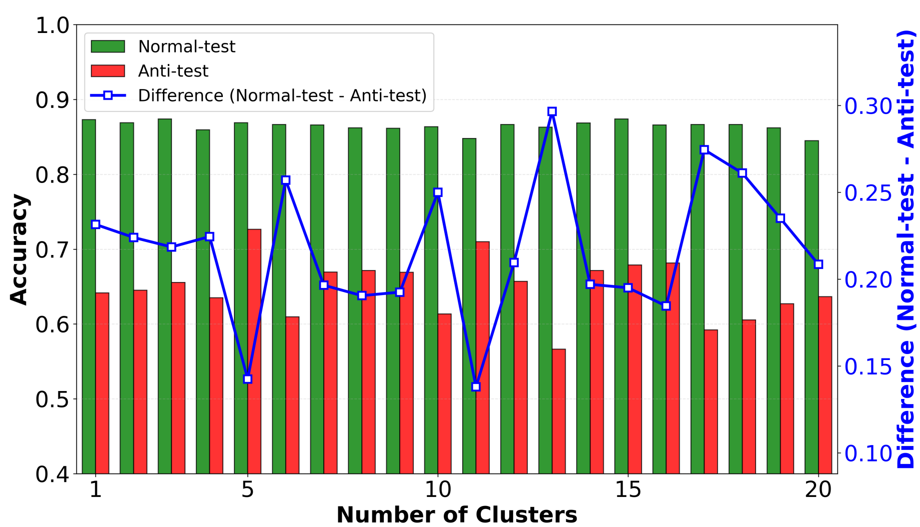

Fig. 12: **Bar Chart of Dataset Experimental Results for Different Numbers of Clusters at an MSFS Embedding Stretching Ratio of 0.01**

Fig. 13: **Bar Chart of Dataset Experimental Results for Different Numbers of Clusters at an MSFS Embedding Stretching Ratio of 0.1**

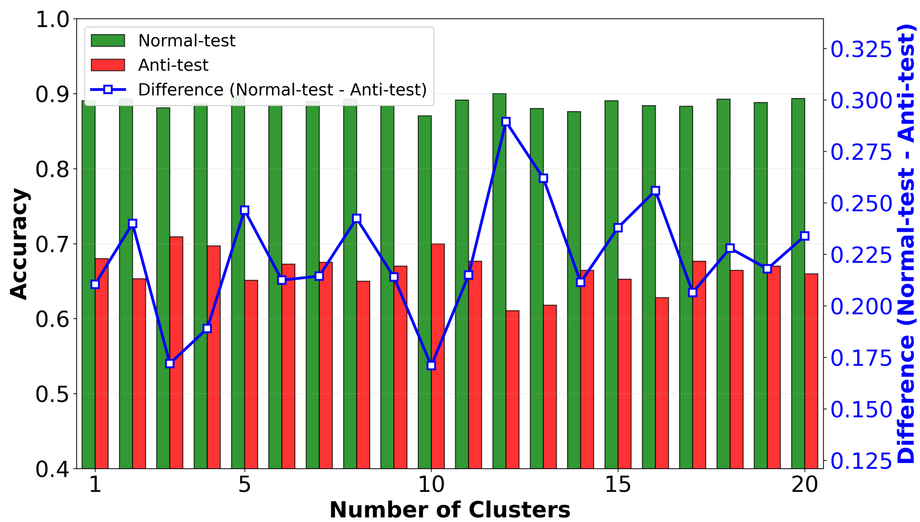

Fig. 14: **Bar Chart of Dataset Experimental Results for Different Numbers of Clusters at an MSFS Embedding Stretching Ratio of 10**

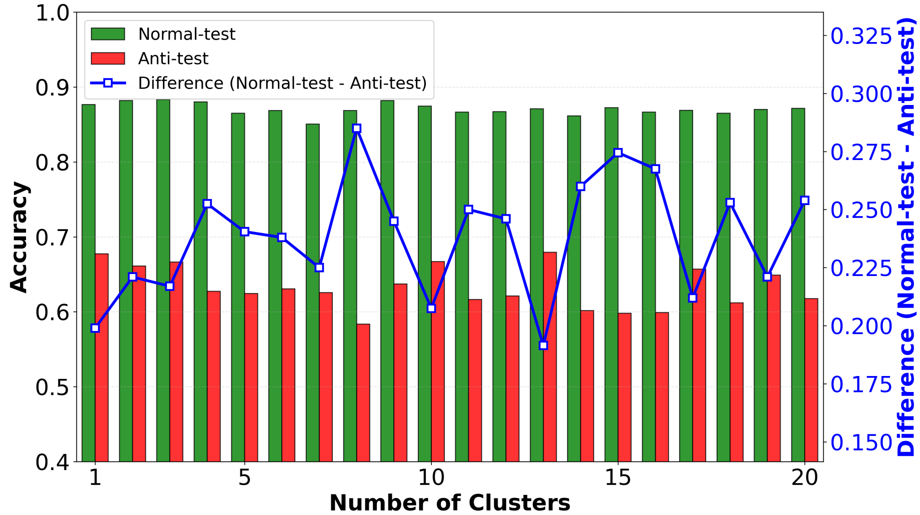

Fig. 15: **Bar Chart of Dataset Experimental Results for Different Numbers of Clusters at an MSFS Embedding Stretching Ratio of 100**

Fig. 16: **Bar Chart of Dataset Experimental Results for Different Numbers of Clusters at an MSFS Embedding Stretching Ratio of 1000**

Fig. 17: **Bar Chart of Dataset Experimental Results for Different Numbers of Clusters at an MSFS Embedding Stretching Ratio of 10000**